

Group 6

Power Cable V2

Final Presentation

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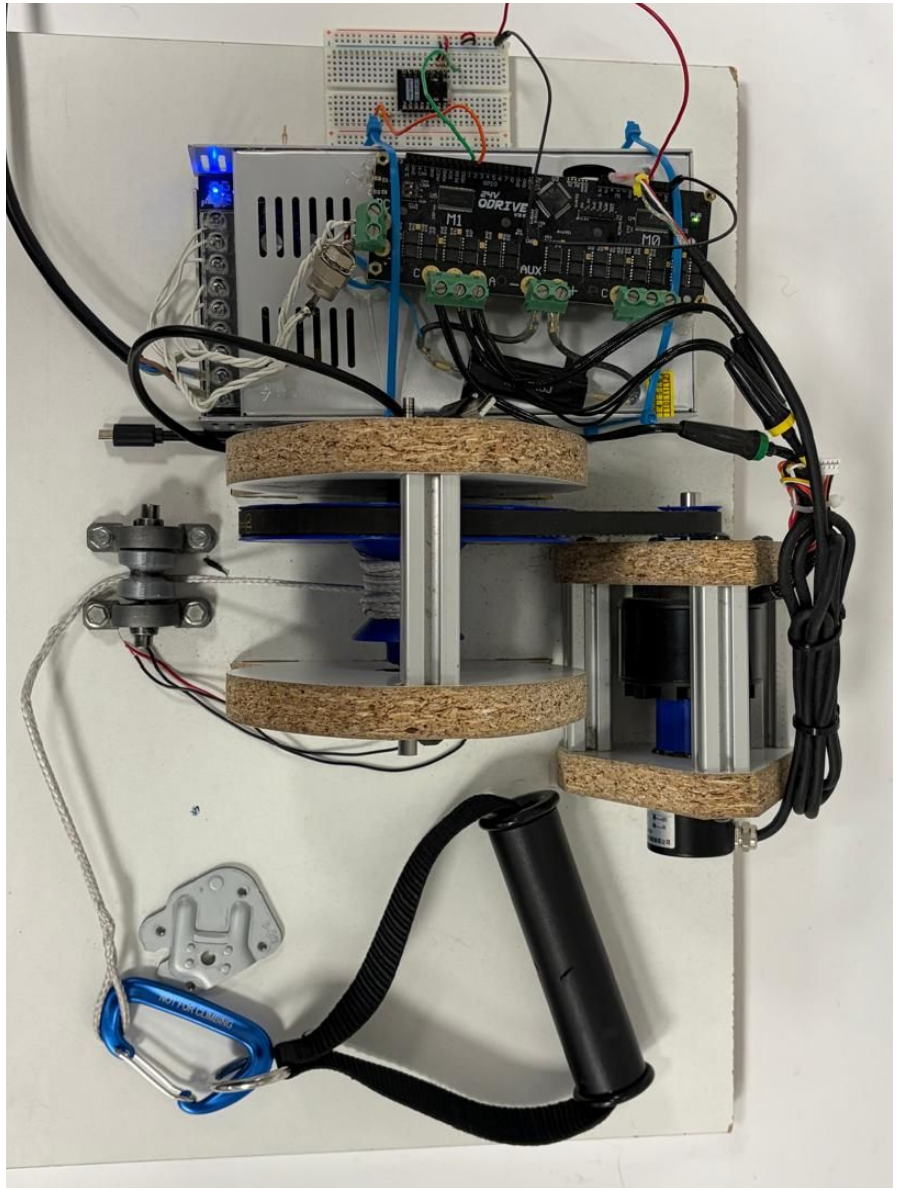
Introduction

- Power Cable: Uses a motorized cable system to provide electronically controlled resistance.

- Drawback: Resistance is fixed at the beginning of each set and cannot be adjusted during exercise.

- Project focus: Develop an intelligent resistance system that automatically adjusts resistance according to the user's current capacity.

- Intends for user favorability for home workouts.



Previous Prototype

Problem Statement

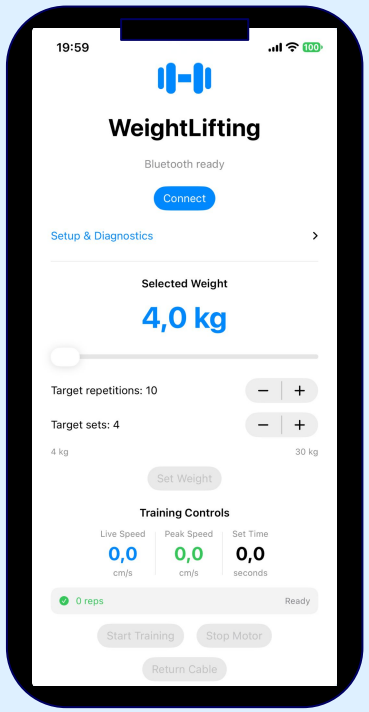
- **Market Limitations:** Home gym equipment is often bulky, expensive (e.g., commercial alternatives costing upwards of 28,300 SEK), and noisy.

- **Technical Drawback:** Fixed-resistance motorized cable devices set load prior to exercise, failing to adjust to daily user strength fluctuations or intra-workout fatigue.

- **Training Inefficiency:** Without dynamic resistance adjustment during reps, users cannot effectively achieve progressive overload or muscular failure for optimal hypertrophy training.

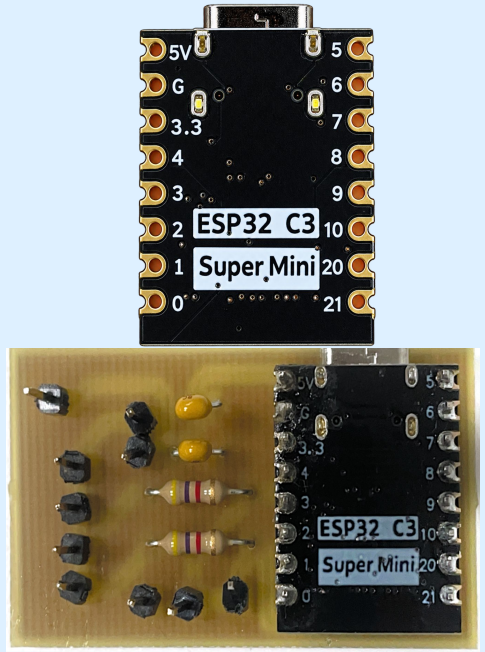
Technical Approach

App/HTML



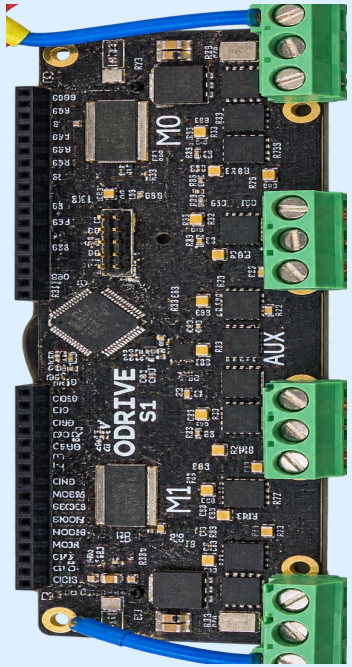
ESP32 C3

Bluetooth bridge



ODrive

motor controller



Motor + Encoder

feedback Kg to current



Cable Mechanism

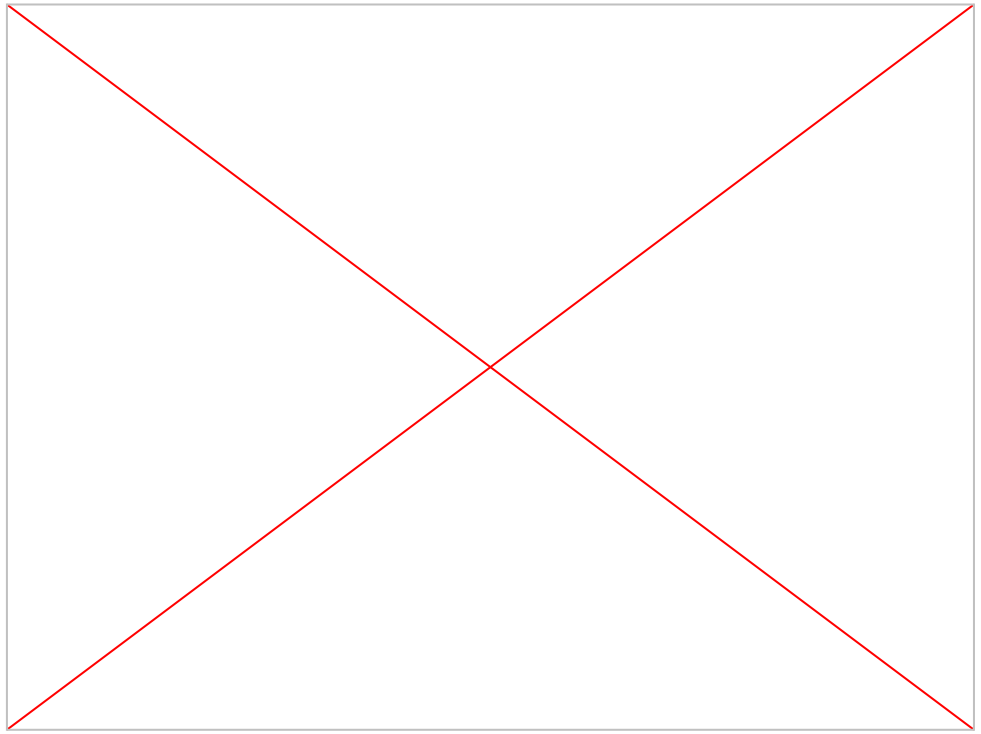
resistance to user

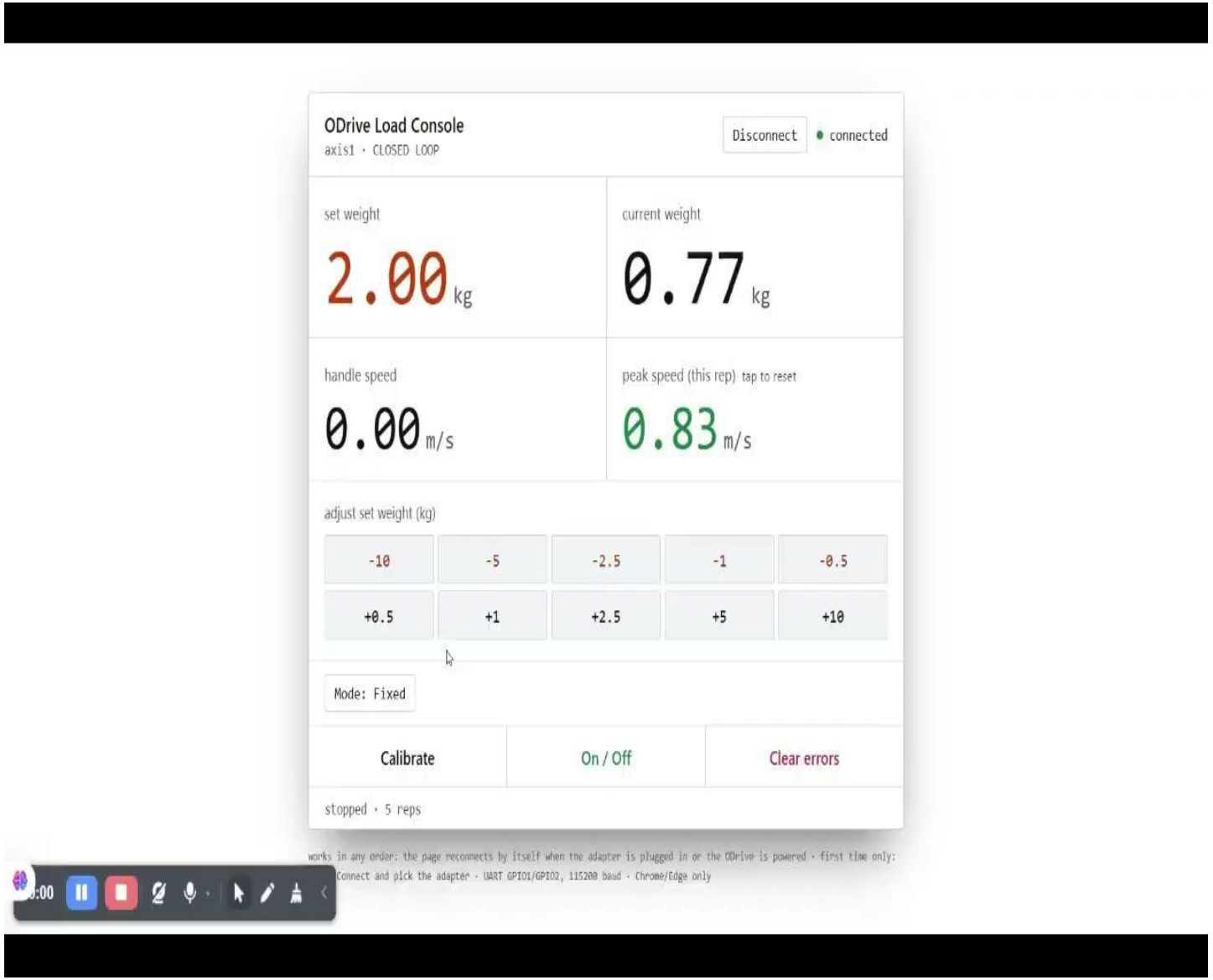


User Interaction with USB

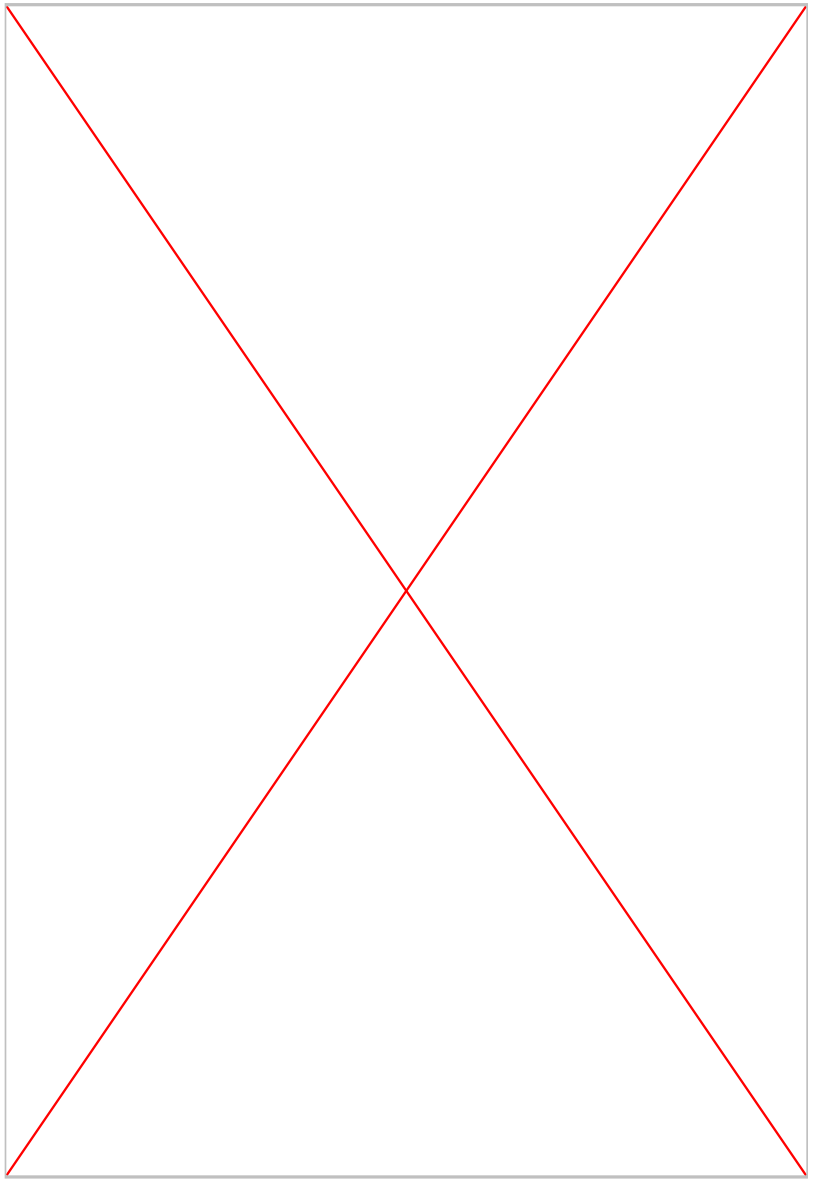
Browser Interface:

- Fixed resistance setting
- Adaptive resistance setting
- Access and operate directly from GitHub
- Live training data

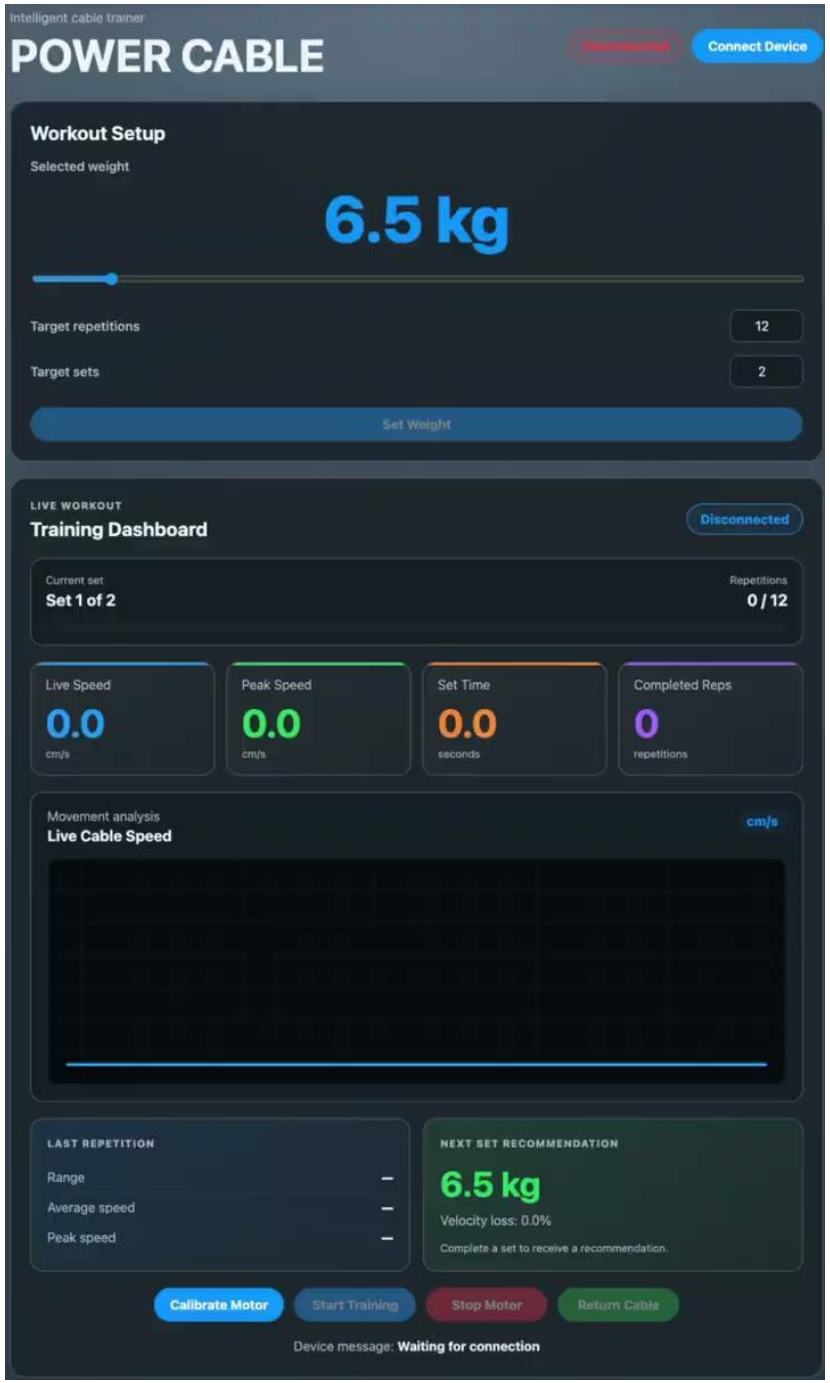




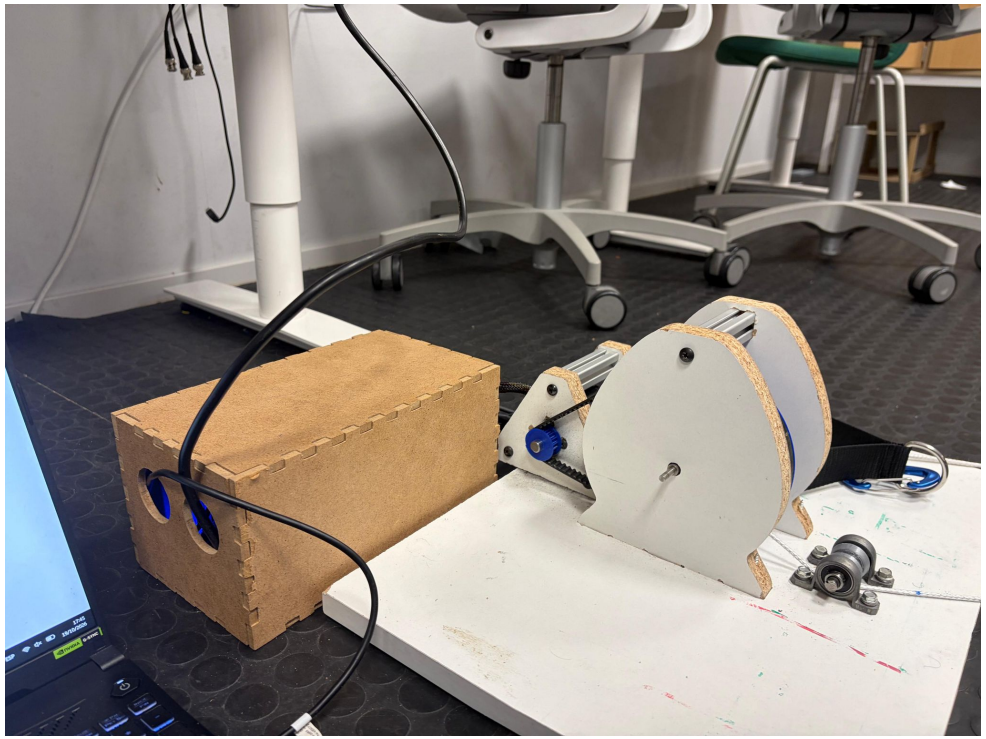
Wireless User Interaction



iOS App - Bluetooth	Web Dashboard - Wi-Fi
portable Workout control	no installation required
Live training data	Live speed visualisation
Sound feedback	Next-set recommendation

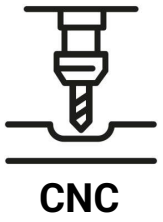


Results



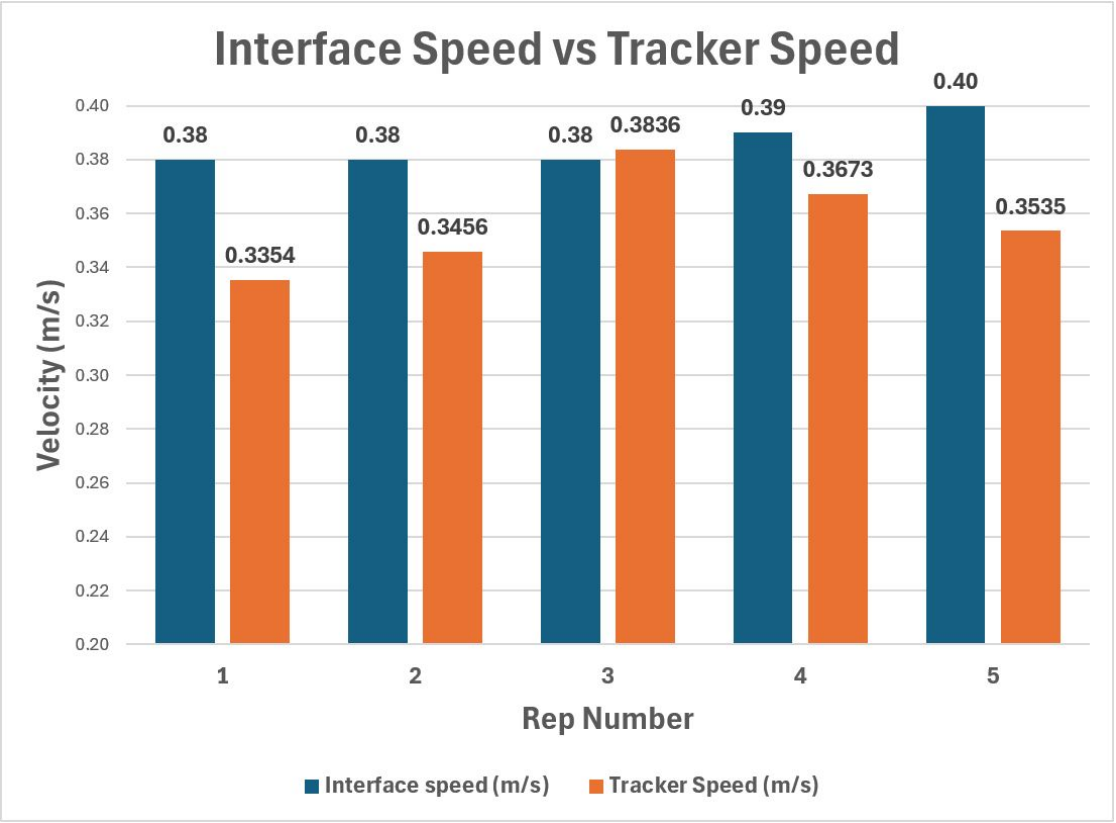
Final Prototype:

- Covers wires and circuits
- Safe and aesthetic
- Built with CNC machining



Evaluation

- Accurate user speed measuring
- 8 % average error



PROJECT METHODOLOGY

	Hardware	HTML Software	Wi-Fi and App Software
Did Well	• Created a closed box to cover the cables from the power supply an ODrive safely.	• Implemented the python project into an HTML interface operable from GitHub • Designed the adaptive resistance mode tracking the rep speed	• Developed the iOS app using Bluetooth • Developed the ESP32-WiFi dashboard • Integrated both interfaces with the shared ESP32 firmware
Would do differently	• Find a better way of anchoring the cable system.	• More research on training efficiency to make the adaptive mode better	• Test the adaptive system with more users • Start complete system testing earlier



Tack for öss

Questions?

POWER CABLE V2